

Unified Intelligence Theory and the Artificial Intelligence Level: A Cumulative, Harness-Measurable Hierarchy

**Five Conceptual Intelligence Families, Six Measurable Coordinates,
a Gated $U0-U\Omega$ Scale, and the AI-Level Score of a Base Model
Across a Standardized Harness Ladder**

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Abstract

Intelligence labels often conflate reasoning, persistent learning, collective organization, autonomous task completion, and evidence-grounded self-modeling. We present a measurement framework with five conceptual families—Cognitive, Individual, Organizational, Delegation, and Self-Awareness—and six directly measured coordinates $[C, I, O, T, H, SA]$, where Delegation is derived from task difficulty T , human cognitive intervention H , and externally verified reliability. Memory Capability M is not a sixth family and not a relabelling of I : it is certified on its own ladder $M0-M\Omega$, each level a witness with a control — recovery with provenance against an ablated-state arm, stale-state suppression under supersession, consolidation that survives hiding the raw episodes, repair against a health oracle the pair cannot write to, cross-project lineage, and a governed change to the memory mechanism itself — and is then joined to I by a one-way prerequisite gate, so that storage never promotes learning. Memory certification is cumulative in capability rather than in implementation, and no level may be skipped. The Unified Intelligence scale $U0-U\Omega$ is cumulative and gated, and the experimental unit is a frozen model–harness pair. GUI/screen understanding enters as a domain of C — witnesses C_k^{GUI} of the existing levels with a diagnostic perceptual subscale $GP0-GP5$ beside them, matched oracle-screen and perfect-actuator arms attributing a failure to perception, cognition or action — so that computer use is measured without a new family or a new ladder. Every ladder is tested in one style, the one in which Individual Intelligence was already written: a construct stated as a contrast, a prerequisite that can block and never promote, a witness with a control arm, a gate that is a product of named factors times lower-level retention, and a certification law. Cumulative structure is typed rather than universal: retention for the capability ladders $C, I, M, O, SA_{1:\Omega}, U$ and the GP diagnostic, a frontier for delegation across T at fixed H and p , nesting for the harness generations, and none for the T and H axes. Levels are stated as invariant operational definitions: each is a contrast that must hold or a structural property of the item, never a difficulty threshold, so that models move through the scale and the scale does not move with the models. On that principle the paper fixes the boundaries the upper levels turn on: Θ , the declared mutable mechanism whose governed change is $I3$; Ψ , the process that proposes such changes, whose own improvement is $I4$; incorporation, which separates $I5$ from remembering a discovery; and reachability, which separates $I\Omega$ from one impressive instrument. Two preliminary programs test the instrument. Across three 48-episode harness curves, a success-only delegation score was invariant to an acceptance-only step for both stronger executors even as each one’s false completions fell from 2/12 to 0/12; a delivered-outcome correction restores sensitivity. In a sealed, post-freeze mechanism probe, one executor passed 5/12 instances versus 1/12